

Hands-on Activity 4.1 Creating a Responsive Portfolio Page with Flexbox and CSS Grid	
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1. Discussion	
This module is all about the usage of Flexbox and Grid system in CSS. This tackles the different properties in dealing with flexbox which is mainly the display, flex direction and justify content. For flexbox it is always dealing with the manipulation of the main axis and cross axis. These two dictate the placement and shift of items in the layout. Then for the Grid System in CSS, this is mainly about using grid-row and grid-column. This system imitates the behavior of a grid where we can shrink or expand the sizes of the boxes in the grid and also the placement of items inside the grid.	
3. Procedure	
Scenario You are creating a responsive portfolio page to showcase your projects. The page will include a header, a navigation bar, a main content area with project cards, and a footer. You'll use Flexbox to create the header and navigation bar, and CSS Grid to layout the project cards in the main content area. We have provided a basic HTML skeleton for your portfolio page in the editor under the index.html file. Your task is to apply the necessary CSS styles to this HTML structure to bring your portfolio to life using Flexbox and CSS Grid. Instructions Follow the instructions below to style the webpage step by step: 1. Apply Flexbox to the Header and Navigation Bar Set the display of the header and nav elements to flex. Justify the content of the header to space between. Align the items in the nav to the center. 2. Style the Navigation Links Set the display of the nav a elements to inline-block. Add padding and margin to the navigation links for spacing. 3. Apply CSS Grid to the Main Content Area Set the display of the .projects container to grid. Define the grid template columns to create a responsive grid layout with three columns. Add gap between the grid items for spacing. 4. Style the Project Cards Add padding, margin, and border to the .project-card elements to make them stand out. Apply box-shadow for a subtle shadow effect. 5. Make the Layout Responsive Use media queries to adjust the grid layout for smaller screens by changing the number of columns. Ensure the navigation bar becomes a column on smaller screens using Flexbox properties.	

6. Review and Test

Review your HTML and CSS to ensure all elements are styled correctly and the CSS rules are applied as expected.

Experiment with different styles and properties to see how they affect your webpage.

4. Output

HTML CODE

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Portfolio</title>
  <link rel="stylesheet" href="output.css">
</head>
<body>
  <header>
    <h1>My Portfolio</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Projects</a>
      <a href="#">Contact</a>
    </nav>
  </header>
  <main>
    <section class="projects">
      <div class="project-card">
        <h2>Project 1</h2>
        <p>Description of project 1.</p>
      </div>
      <div class="project-card">
        <h2>Project 2</h2>
        <p>Description of project 2.</p>
      </div>
      <div class="project-card">
        <h2>Project 3</h2>
        <p>Description of project 3.</p>
      </div>
    </section>
  </main>
  <footer>
    <p>© 2024 My Portfolio</p>
  </footer>
</body>
</html>
```

CSS CODE

```
CSS
/* Base styles */
body {
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 0;
}

header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background-color: #333;
  color: white;
  padding: 10px 20px;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 0 10px;
}

nav a:hover {
  text-decoration: underline;
}

.projects {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 20px;
  padding: 20px;
}

.project-card {
  background-color: #f4f4f4;
  padding: 20px;
  border: 1px solid #ddd;
  box-shadow: 2px 2px 12px rgba(0,0,0,0.1);
}

@media (max-width: 768px) {
  .projects {
    grid-template-columns: 1fr;
  }

  nav {
    flex-direction: column;
  }
}
```

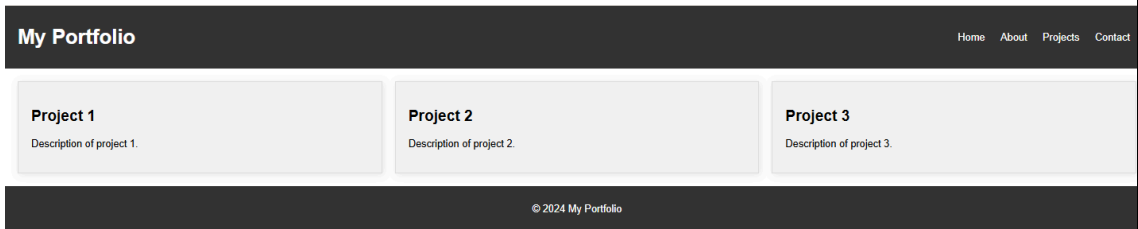
```

        align-items: flex-start;
    }

    nav a {
        margin: 5px 0;
    }
}

footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 10px 20px;
}

```



5. Supplementary Activity

HTML CODE

```

HTML
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Portfolio</title>
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <!-- Archivo Font -->
    <link
href="https://fonts.googleapis.com/css2?family=Archivo:wght@300;400;600;
700&display=swap" rel="stylesheet">

    <link rel="stylesheet" href="supple.css">

```

```

</head>
<body>

    <header class="navbar">
        <h1 class="logo">THE PORTFOLIO</h1>
        <nav>
            <a href="#hero">Home</a>
            <a href="#projects">Projects</a>
            <a href="#contact">Contact</a>
        </nav>
    </header>

    <section id="hero" class="hero">
        <div class="hero-text">
            <h2>Hello, I'm Jhon Hendricks Bautista</h2>
            <p>
                I am a 3rd yr BSCPE student, currently taking the Data
                Science Track Elective.
                I am a passionate developer building intelligent
                systems,
                computer vision projects, and modern web applications.
            </p>
            <a href="#projects" class="btn">View My Work</a>
        </div>

        <div class="hero-image">
            
        </div>
    </section>

    <!-- PROJECTS -->
    <section id="projects" class="projects-section">
        <h2 class="section-title">My Projects</h2>

        <div class="projects-grid">
            <div class="project-card">
                <h3>Water Hyacinth Detection</h3>
                <p>Object detection model for detecting invasive
                plants.</p>
                <div class="card-links">
                    <a
                href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Git
                Hub</a>

                    <a
                href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Det
                ails</a>

                </div>
            </div>

            <div class="project-card">

```

```

        <h3>Brake Fade Predictor</h3>
        <p>Machine learning model to predict brake fade in
vehicles.</p>
        <div class="card-links">
            <a
href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Git
Hub</a>
            <a
href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Det
ails</a>
        </div>
    </div>

    <div class="project-card">
        <h3>Cooking Oil Disposal System</h3>
        <p>Tech solution for managing improper cooking oil
disposal.</p>
        <div class="card-links">
            <a
href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Git
Hub</a>
            <a
href="https://github.com/JhonHendricksBautista/CPE313_Bautista-Jhon">Det
ails</a>
        </div>
    </div>
</div>
</section>

<footer>
    <p>© 2026 Jhon Bautista | All Rights Reserved</p>
</footer>

</body>
</html>

```

CSS CODE

```

CSS
/* ===== BASE ===== */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

```

```
body {
  font-family: 'Archivo', sans-serif;
  background-color: #0F172A;
  color: #E2E8F0;
  line-height: 1.6;
}

/* ===== NAVBAR ===== */
.navbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px 8%;
  background-color: #1E293B;
  position: sticky;
  top: 0;
}

.logo {
  font-weight: 700;
  color: #3B82F6;
}

nav a {
  color: #E2E8F0;
  text-decoration: none;
  margin-left: 25px;
  font-weight: 500;
  transition: 0.3s;
}

nav a:hover {
  color: #3B82F6;
}

.hero {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 100px 8%;
  gap: 40px;
  flex-wrap: wrap;
}

.hero-text {
  flex: 1;
}
```

```
.hero-text h2 {
  font-size: 2.8rem;
  color: #3B82F6;
  margin-bottom: 20px;
}

.hero-text p {
  margin-bottom: 25px;
  max-width: 500px;
  color: #CBD5E1;
}

.btn {
  display: inline-block;
  padding: 12px 24px;
  background-color: #1D4ED8;
  color: white;
  text-decoration: none;
  border-radius: 6px;
  transition: 0.3s;
}

.btn:hover {
  background-color: #3B82F6;
}

.hero-image img {
  width: 280px;
  border-radius: 50%;
  border: 5px solid #1D4ED8;
}

.projects-section {
  padding: 100px 8%;
  background-color: #1E293B;
}

.section-title {
  text-align: center;
  margin-bottom: 60px;
  color: #3B82F6;
  font-size: 2rem;
}

.projects-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
  gap: 30px;
```



```
}

.project-card {
  background-color: #0F172A;
  padding: 30px;
  border-radius: 12px;
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  border: 1px solid #334155;
  transition: 0.3s;
}

.project-card h3 {
  color: #3B82F6;
  margin-bottom: 15px;
}

.project-card p {
  color: #94A3B8;
}

.card-links {
  margin-top: 25px;
  display: flex;
  justify-content: space-between;
}

.card-links a {
  text-decoration: none;
  color: #1D4ED8;
  font-weight: 600;
}

.card-links a:hover {
  color: #3B82F6;
}

footer {
  background-color: #002157;
  color: #fcfcfc;
  text-align: center;
  padding: 20px;
}

@media (max-width: 768px) {
```

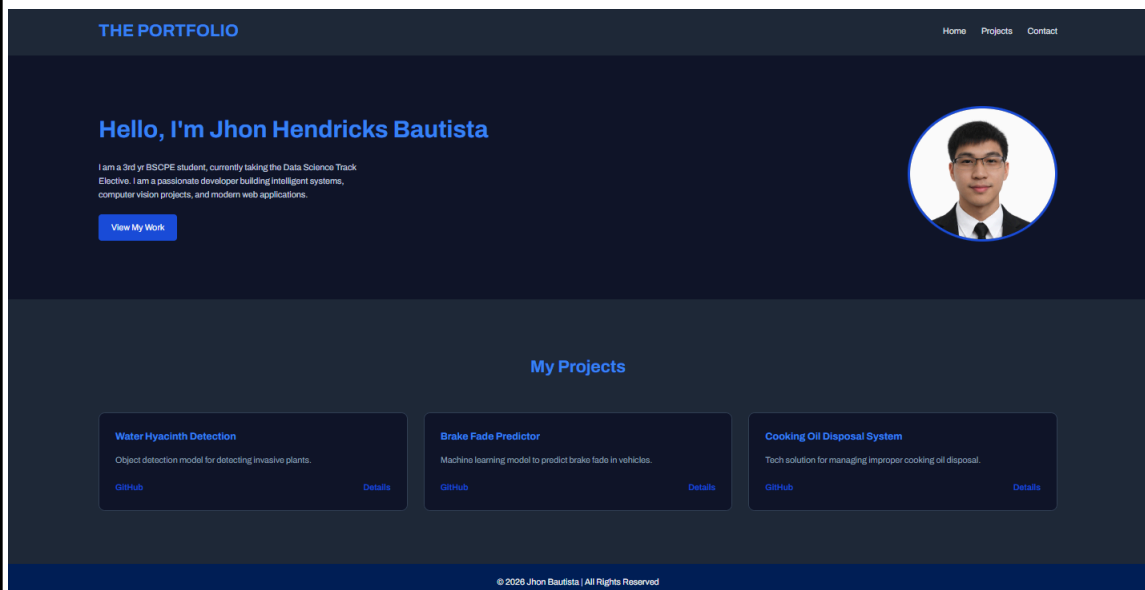
```

.hero {
  flex-direction: column;
  text-align: center;
}

.hero-image img {
  margin-top: 30px;
}

.navbar {
  flex-direction: column;
  gap: 10px;
}
}

```



CONCLUSION

After performing this activity I was able to learn about the use of another layout system in CSS. I learned about the Flexbox and Grid system in CSS. I learned the Flexbox is a better approach than floats since it works more automatically and fluidly without too much modification needed. I learned that the CSS grid system works like building a table or boxes wherein we can control the sizes of the boxes. After learning these two layouting system, I used it in a portfolio website wherein I was able to effectively space the different elements out properly using the flexbox. Then for the project cards I used the grid so that they would have uniform spacing and sizes.